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ABSTRACT

A plasma device has been created to study dynamic plasma coupling in an $E \times B$ -drifting magnetized plasma. The $E \times B$ magnetized plasma device is a 1.2 m diameter by 2 m long cylindrical chamber with two sets of Helmholtz coils in a mirror configuration. A steady-state axial hollow cathode source injects a plasma discharge in electrical contact with a floating conductor at a range that forms a unique axisymmetric equipotential surface or Virtual Cathode Lightsaber (VCL). The VCL generates two plasma populations streaming relative to one another providing a suitable environment for the investigation of dynamic plasma coupling. The plasma density, radial electric field, and plasma rotational velocity outside the VCL are shown to be influenced by the current-voltage relationship of the cathode and applied magnetic field strength. A basic characterization of the device and plasma environment is presented with an emphasis on diagnostics systems and the analytical techniques utilized.

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I. INTRODUCTION

Dynamic plasma coupling or momentum-coupling in $E \times B$ drifting magnetized plasma spans a broad range of phenomena, including coupling between the magnetosphere and ionosphere,¹ Alfvén's critical ionization velocity² (CIV), plasma injection experiments in the ionosphere,^{3,4} centrifugal plasma confinement,⁵ charged and/or large extended spacecraft including electrodynamic tether systems,⁶ and plasma expansion associated with high altitude nuclear explosion (HANE).^{7,8} In these areas and more, there exist certain knowledge gaps that may be investigated by the creation of a scaled laboratory plasma experiment.

Numerous crossed-field devices⁹ spanning over half a century have been built to explore a variety of different branches in plasma physics. Many notable rotating plasma experiments have focused on thermonuclear fusion and other high-energy density applications, such as the Ixion Experiment at Los Alamos National Laboratories that underwent several upgrades and reported rotational speeds of

80–130 km/s.^{10,11} Rotational plasmas have been investigated at the Mega Ampere Generator for Plasma Implosion Experiments (MAGPIE) facility at the Imperial College. In 2017, Thompson scattering was used on MAGPIE to determine the rotational velocity of a lab-made accretion disk to be near 60 km/s; the reported electron density and ion temperature were 10^{19} cm^{-3} and $T_i \approx 60 \text{ eV}$ respectively.¹² Japan's GAMMA10 aimed to investigate plasma confinement and heating techniques. The rotational velocities in helium were measured at $v_\phi = 10\text{--}30 \text{ km/s}$. The ion density (n_i), electron temperature (T_e), ion temperature (T_i), and magnetic field (B) were $n_i = 2\text{--}4.5 \times 10^{14} \text{ cm}^{-3}$, $T_e = 5\text{--}8 \text{ eV}$, $T_i = 9\text{--}15 \text{ eV}$, and $B = 500\text{--}870 \text{ G}$, respectively.¹³ The University of Maryland built the Maryland Centrifugal eXperiment (MCX) in 2003. The $E \times B$ fields produced a rotating plasma with reported parameters of $n_i \approx 5 \times 10^{14} \text{ cm}^{-3}$, $T_e \approx 100 \text{ eV}$,¹⁴ and $T_i \approx 150 \text{ eV}$.¹⁵ Rotational speeds in hydrogen were reported at 150 km/s at the midplane and 50 km/s at the mirror.¹⁶

While the exploration of rotating plasmas has a rich history, there is still a need for rotating plasma devices for basic research and special technical applications. For the purposes of studying Earth and solar plasma environments, a low- β type A device⁹ that produces a quasi-stationary, rotating plasma using the crossed field technique, has been developed at the Air Force Research Laboratory - Space Vehicles Directorate (RV) in Albuquerque NM. The $E \times B$ magnetized plasma device (EMPD) drives a rotating plasma by a sustained discharge nicknamed the “Virtual Cathode Lightsaber” (VCL) shown in Fig. 1(b). The VCL is generated by a hollow cathode (HC) oriented along the chamber-magnetic field axis. The magnetic field is created by an external set of Helmholtz coils with an additional set of end coils to produce a mirror configuration. The VCL electron beam is reflected by a floating metal plate or repeller¹⁷ at the far end of the chamber, thus trapping electrons and improving

uniformity. Glass end doors force the chamber walls to become the anode. After subsequent reflections, the beam-discharge forms an ionization induced plasma column and approximate equipotential.

The VCL establishes a transverse radially inward electric field causing the partially ionized plasma outside the VCL to rotate. A condition of iso-rotation for axisymmetric fields (or rigid rotor approximation)¹⁸ may not always apply because of the significant neutral density in the rotating plasma volume.⁹ The VCL also creates two distinct plasma populations; one in the VCL and one in the rotational volume. The study of the hollow cathode itself in this environment provides insight into the behavior of plasma contactors.^{19–21} The interaction of the two coupled co-moving plasma populations may be influenced via the driving cathode potential and applied field. This creates a suitable environment to investigate dynamic plasma coupling, ion-neutral interactions and instabilities, cross-

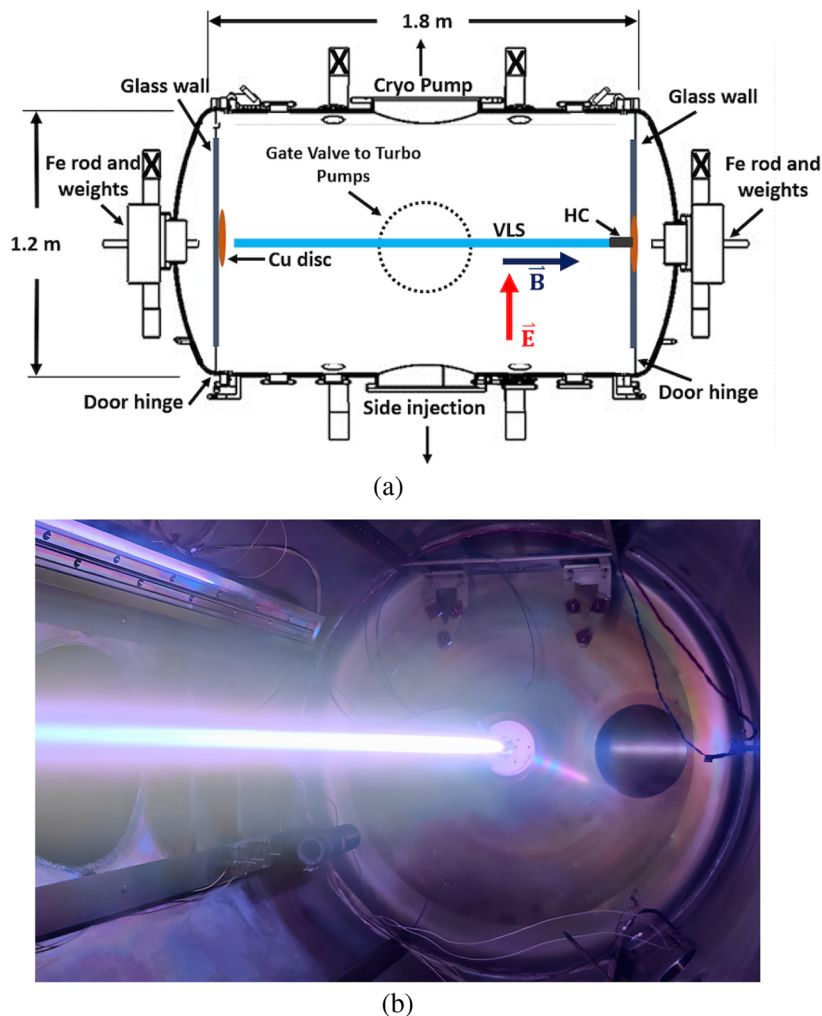


FIG. 1. (a) Top-down cross section of EMPD highlighting essential components used during operation. The electric and magnetic field vectors, HC location, copper disks, glass walls, and electromagnets with iron cores are labeled. The midplane coincident with all the plasma measurements is indicated by the dashed line. (b) VCL operation with the hollow cathode in jet mode with $I_c = 20$ A at 20 SCCM.

field particle transport, velocity-limiting processes, and rotational shear.

One device²² used a similar magnetically confined hollow cathode electron source in a medium pressure Ar background (10^{-4} – 10^{-3} Torr) and observed a rotating plasma with a peculiar “vane” structure. That device simply dumped the electron beam into an anode on the beam axis. They do not report how they handled the heating or how uniform the beam potential was along the axis. In the case of EMPD, the VCL operates with up to 20 kW of electron beam power, which caused a serious problem when briefly operated with an adsorbing on-axis anode. EMPD uses an electrically floating target that charges and reflects the electron beam thus contributing to potential uniformity and perhaps suppressing any vanes (we have observed none). The target does not significantly heat and is suspended by a common insulated copper wire. EMPD also employs borosilicate glass end “doors,” which support a transverse radial electric field. Together with the floating target, EMPD forces the outer walls to become the anode. These features along with centrally located pumping ports help EMPD achieve axially uniform $E \times B$ drifting plasma, which is the objective.

One phenomenon we are investigating is any limits on plasma rotation. Magnetohydrodynamic (MHD) theory dictates a hard velocity limit at the Alfvén velocity, $v_A = B(\mu_0\rho)^{-1/2}$, where ρ is the plasma volumetric mass density and μ_0 is the permeability of free space. A soft velocity limit in the presence of a significant neutral population is the critical ionization velocity,²³

$$v_{crit} = \sqrt{\frac{2e\phi_i}{m_i}}. \quad (1)$$

Here, e/m_i is the charge-to-mass ratio of the ion species and ϕ_i is the ionization energy of the neutral gas species. The CIV effect acts as an angular momentum sink in a rotating plasma by shifting the momentum from ions to the ionization of neutrals, while raising the ion density and, consequently, the moment of inertia in the rotational volume. The process manifests as a limit on the attainable radial voltage gradient, which is responsible for driving the rotation until the neutral population is exhausted.

The physical mechanism driving CIV is not completely understood, but it is generally thought to be the lower hybrid instability.²⁴ However, many uncertainties remain, such as the requirements for the neutral density.

Another research aim for EMPD consists of electrostatic probes in flowing magnetized plasma. There is a large body of theory and best practices for Langmuir probes.^{25,26} They are generally well understood except for the case of the positive probe in a flowing magnetized plasma. The few experiments highlighting this problem show anomalously high currents that exceed the Parker–Murphy limit; a limit based on the conservation of canonical angular momentum.^{27,28} Despite this, Langmuir probes are still used in flowing magnetized plasma regularly, such as in the field of electric propulsion. Efforts have been underway toward building a theoretical framework and computational models^{6,29} for a positive probe under these plasma conditions but verification in the lab is still needed.

II. EXPERIMENTAL APPARATUS

EMPD is located within the Spacecraft Charging and Instrument Calibration Laboratory (SCICL) of the Air Force Research Laboratory at Kirtland Air Force Base. A top-down cross section of EMPD is shown in Fig. 1(a). The 304 stainless steel cylindrical vacuum vessel has a 1.2 m inner diameter and is 1.8 m long. The chamber is equipped with two hinged access doors and numerous ports for diagnostics, power/gas feedthroughs, and pumping. Three dry scroll pumps are used to bring the chamber to a rough vacuum before two turbo-molecular pumps and a cryogenic pump initialize a baseline pressure of ~ 6 – $8 \mu\text{Torr}$ as measured by a nitrogen-calibrated ionization gauge.

The magnetic field is generated by four electromagnetic coils from Alpha Magnetics and Coils, Inc. The center coils are 64 in. (1.6 m) ID $\times$ 74 in. (1.9 m) OD, water cooled with 54 turns each, and placed in a Helmholtz configuration. The end coils are 24 in. (0.6 m) ID $\times$ 34 in. (0.9 m) OD, water cooled, with 60 turns each. The turns are grouped in two series circuits each for the end coils and three circuits for the center coils for a total of ten circuits of similar resistance, powered in parallel pairs for a max current close to 500 A to each turn. The coils are powered by five 10 kW TDK-Lambda GEN 3U 1000 A DC power supplies. To enhance the mirror ratio and shape the magnetic field in the end pinch, the chamber doors each have a pure iron 1 in. (25.4 cm) rod welded through an on-axis flange providing an interior and exterior rod or “core,” with a pure iron Olympic barbell adapter, outside which multiple 35 lb cast iron weights can be added. The resultant magnetic field, shown in Fig. 2, was produced with a finite element method (FEM),³⁰ and verified with a Lake Shore F71 Cryogenics Tesla Meter with a measured error of at most 2.5% on axis when approaching the HC location. $B_{\min}$ is on the axial midplane, and $B_{\max}$ is at the HC location. The coil current was selected to generate the magnetic field $B_{\max} = 189$ G and mirror ratio $R_m = B_{\max}/B_{\min} = 1.67$. The area of high field (magenta) near the chamber axis is the iron magnet core that enhances the pinch field near $B_{\max}$. The FEM model calculates a mirror ratio enhancement of 27% with installation of the cores. For peak current on all the coils, the field at $B_{\max}$ can reach 450 G and the mirror ratio for operation is typically chosen to be within $R_m = 1.6$ – 3.6 as higher values decrease the uniform region of the axial mirror profile.

A. Hollow cathode plasma source description

The VCL discharge is generated by a Plasma Controls H6 model heaterless hollow cathode, as shown in Fig. 3, with a 1 mm diameter aperture and 6.35 mm OD keeper. The replaceable keeper aperture plate is held in place by a threaded cap. Heaterless cathodes possess several beneficial qualities, such as minimal peripheral equipment needed for operation, long operational lifetime, and a simplified start-up procedure.^{31–33}

Hollow cathode ignition begins by feeding neutral gas to the device. A potential bias of 600 V is applied to the keeper with respect to the cathode tube, initiating a Paschen discharge. If a discharge does not occur, closing a solenoid switch in the gas feed line allows for a gas build up and subsequent pulse that facilitates dielectric breakdown of the gas. As electrons are attracted to the keeper, ion bombardment rapidly heats the cathode tube, which indirectly heats an internal low work function thermionic emitter that initiates the

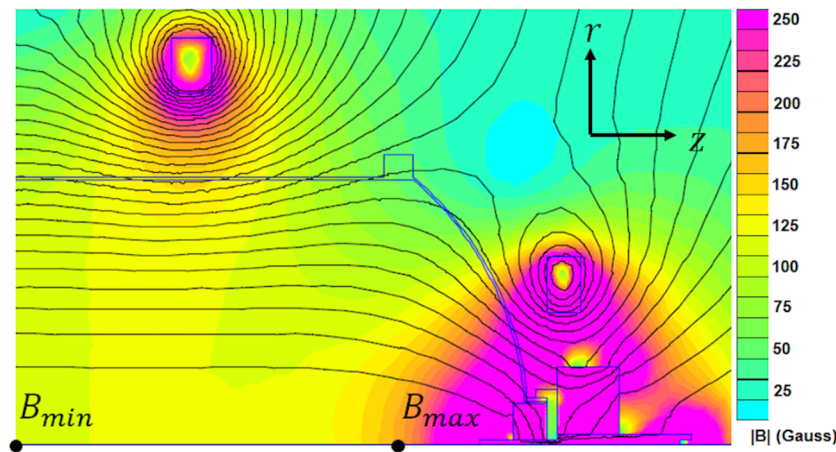


FIG. 2. FEM contour map of magnetic flux density for currents corresponding to the $B_{\max} = 189$ G and $R_m = 1.67$ configuration. $B_{\min}$ is coincident with the axis and chamber midplane. $B_{\max}$ is at the HC location. Both central and end coils with the concentric iron magnet core are visible.

electron discharge. A small region of high ionization close to the cathode orifice provides space charge neutralization of the electron discharge. Typically, argon is used in EMPD experiments with mass flow rates between 8 and 30 SCCM, although the HC has been operated successfully with argon and neon mixtures of up to 90% neon.

During operation, the cathode discharge current (I_c) is limited between 2 and 30 A by the power supply, which is provided by a 20 kW Magna-power power supply (300 V, 60 A). The VCL is typically operated at less than 30 A. This corresponds to a cathode voltage (V_c) in the range of 50–240 V below ground potential depending on the chamber pressure, gas flow rate, and applied magnetic field. Several current–voltage (I – V) characteristic curves are shown in Fig. 4 with varying magnetic field strengths. The keeper voltages over this range are approximately constant at 20 V (1 A) with respect to the cathode emission tube.

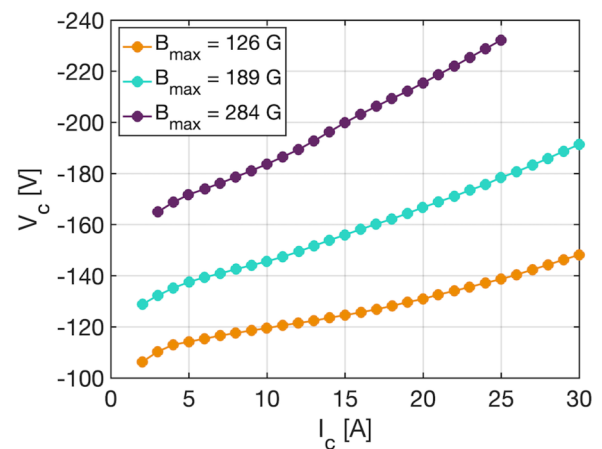


FIG. 4. HC I – V characteristic curves for three magnetic field strengths. For all cases, the mirror ratio was constant at $R_m = 1.67$, the gas fill pressure was $62 \mu\text{Torr}$ at 20 SCCM Ar gas flow rate, and the keeper was held at 1.1 A and 20 V. The standard deviation of the voltage is at most 1.5 V for all the measurements.

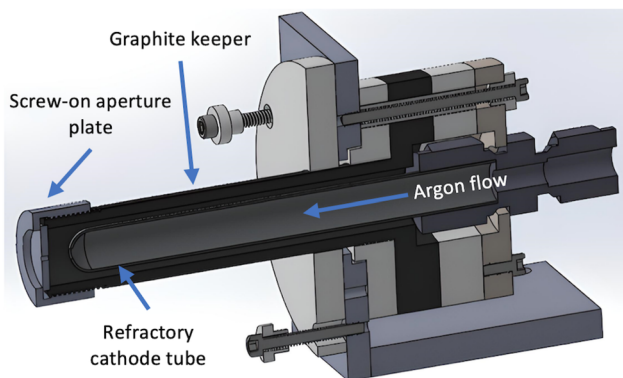


FIG. 3. Cross-section of an H6 HC assembly fabricated by Plasma Controls Ltd.

B. Plasma characterization

In general, virtual cathodes rely on an electron beam reflecting at a stagnation point after a transitory region, oftentimes a screen mesh, to help build up space-charge at the opposite end of the beam. EMPD's HC is mounted at the center of a 15 cm diameter carbon (Aquadag) coated copper back-plate biased at the keeper potential and installed into the center of an insulating 96.5 cm diameter borosilicate glass wall. An identical glass wall at the opposite end of the chamber supports an electrically floating carbon–copper plate. The negatively biased HC device outputs a plasma plume, which then reflects off the floating plate. The conductivity along

the magnetic field between the two plates is much larger than the Pedersen conductivity and establishes an approximate-equipotential at the keeper voltage in this region. The voltage drop between the carbon–copper plates is about 2 V along the VCL. The two glass walls are responsible for preventing current closure between field lines and support the radial electric field in the chamber.

The plasma volume is separated into two regions of interest: the volume inside the VCL where the radial electric field is assumed to be negligible and the region outside the VCL in contact with the glass walls referred to as the rotational plasma volume. The annular rotational volume is axially bound between the two glass doors and radially bound between the inner and outer flux surfaces that terminate on the glass but do not intersect the chamber wall. The inner flux surface is the boundary of the carbon–copper disks and has a radial distance from the center axis of $r = 7.5$ cm. The outer flux surface, as determined from the FEM model, is the outer edge of the glass wall at $r = 48.2$ cm.

EMPD features plasma diagnostics including a radially motorized Langmuir probe, an emissive probe array, and a modified Mach probe/potential analyzer that each sequentially measures the plasma environment. Four plasma environments are analyzed in this study to explore the low to medium energy plasma environment produced by EMPD. These four environments correspond to two different magnetic field configurations, each with two different cathode emission currents. The magnetic field configurations are $B_{\max} = 126$ G and $B_{\max} = 189$ G both with $R_m = 1.67$. The two cathode currents are $I_c = 3$ A and $I_c = 12$ A. All cases were taken on the chamber midplane [dashed line in Fig. 1(a)] with approximate pressure 55 μ Torr and argon gas flow rate of 20 SCCM.

The combined measurements outlined in the following sections are used to calculate a range of plasma parameters inside the rotational volume that are presented in Table I. The measured densities and temperatures are then extrapolated over a typical operational range of neutral pressures $P = 55$ –200 μ Torr and magnetic field strengths $B = 70$ –300 G presented in Table I. The chamber pressure range is determined by the use of the cryogenic pump alongside the turbomolecular pumps at a constant argon gas flow rate of 20 SCCM. The ion temperature is not yet well characterized during plasma rotation. This is subject to on-going work using the RMP (see Sec. II B 3). The notation used for the electron, ion, and neutral parameters (subscript e, i, n , respectively) that have not been defined previously include the neutral number density n_n , the plasma potential V_p , ion sound speed c_s , rotational plasma volume V , Debye length λ_d , electron plasma frequency f_{pe} , cyclotron frequencies f_{ci} and f_{ce} , ion–ion collision frequency ν_{ii} , ion–neutral collision frequency ν_{in} , ion–neutral resonant charge exchange frequency ν_{ex} , the mean free paths λ_i and λ_e , the Pederson (σ_P) and Hall (σ_H) conductivities,³⁴ and the plasma β .

1. Radial Langmuir probe

Langmuir probe measurements were taken using an ALP-150 (Impedans Ltd.) probe and source-measure unit (SMU). The probe is comprised of a 0.5 mm OD tungsten tip affixed to a 6.3 mm OD alumina shaft with its position controlled by a 1.4 mm linear drive stage mounted along the chamber midplane. The probe is capable of taking measurements 10 cm off the chamber wall to several centimeters past the chamber axis. The unit also makes use of several accuracy improving methods, such as a DC compensation

TABLE I. Range of measured plasma parameters for $I_c = 3$ –12 A, $R_m = 1.67$, and 20 SCCM argon gas flow rate in the rotational volume ($7.5 \text{ cm} < r < 48.2 \text{ cm}$). The parameters are extrapolated to operating conditions given by $P = 55$ –200 μ Torr and $B = 70$ –300 G to estimate various other quantities in the volume.

Parameters	Range
$n_i \text{ (cm}^{-3}\text{)}$	$5 \times (10^{10} - 10^{12})$
$n_n \text{ (cm}^{-3}\text{)}$	$(1.8 - 6.4) \times 10^{12}$
$V_p \text{ (V)}$	95–19
$T_e \text{ (eV)}$	6–9
$T_i \text{ (eV)}$	1–2
$c_s \text{ (km/s)}$	4.7–6.0
$v_A \text{ (km/s)}$	46–108
$V \text{ (m}^3\text{)}$	0.95
$\lambda_d \text{ (}\mu\text{m)}$	8–100
$f_{pe} \text{ (GHz)}$	2–20
$f_{ci} \text{ (kHz)}$	2.7–11.5
$f_{ce} \text{ (GHz)}$	0.20–0.84
$\nu_{ii} \text{ (kHz)}$	1.6–325
$\nu_{in} \text{ (Hz)}$	117–603
$\nu_{ex} \text{ (kHz)}$	2.8–14.5
$\lambda_i \text{ (m)}$	0.004–1.05
$\lambda_e \text{ (m)}$	0.07–8.3
$\sigma_P \text{ (S/m)}$	9×10^{-4} –8
$\sigma_H \text{ (S/m)}$	1×10^{-6} –0.3
β	0.006–0.9

algorithm, plasma-ground sheath resistance, and probe tip cleaning function.³⁵

Non-Maxwellian plasma distributions impact cold probes because the voltage is floating with respect to the plasma, and the probe must charge negatively into the electron tail to achieve current balance. For this reason, the transition from the electron retarding region to the electron saturation region is sometimes not well defined. Figure 5 shows an example of a curve near the wall (38 cm), closer to the VCL (22 cm), and near the edge of the VCL (14 cm). The 14 cm curve is truncated due to the power absorbing limits of the probe system.

When the ALPS system cannot report density from the I–V curve, we can estimate the density from the ion collection assuming a cold ($T_i/T_e < 1$) collisionless ($r_p/\lambda_i \ll 1$) plasma and the thin sheath approximation ($r_p/\lambda_d \gg 1$), where the sheath area is assumed to be equal to the probe area ($A_s \approx A_p$). The ion density may then be calculated by setting the ion saturation current at the sheath edge $I_{i,s}$ equal to the Bohm current.²⁵ The equation for ion density is then given by

$$n_i = \frac{2I_{i,s}}{eA_p} \sqrt{\frac{m_i}{k_B T_e}}, \quad (2)$$

where k_B is the Boltzmann constant.

Taking the Bohm current to be equal to the electron current (the product of the Boltzmann factor and electron saturation current), V_p may be determined from the floating voltage V_f ,

$$V_p = V_f + \frac{k_B T_e}{2e} \ln \left(\frac{2m_i}{\pi m_e} \right). \quad (3)$$

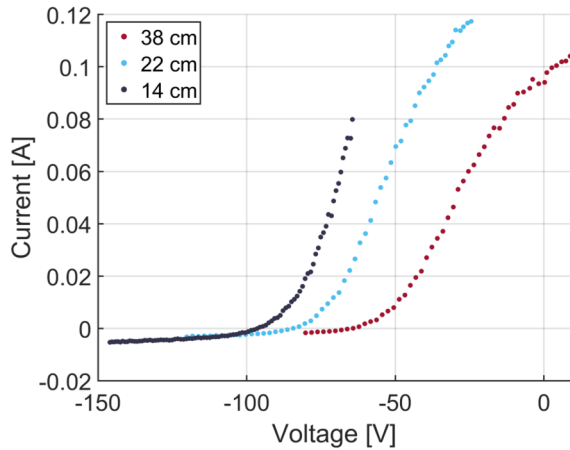


FIG. 5. Langmuir probe I-V curves at three different radial positions with $V_c = 152$ V, $I_c = 12$ A, $V_{keeper} = 16.8$ V, $I_{keeper} = 1$ A, $p = 5 \times 10^{-5}$ Torr, $B_{max} = 140$ G, and $B_{min} = 130$ G. The legend corresponds to the distance from the axis.

This expression is not accurate near the axis (VCL) where the plasma is dominated by high-energy electrons but is sufficient in the further reaches of the plasma, which is our present concern. This is the same effect shown in the I-V curves in Fig. 5; when estimating the plasma potential, the I-V “knee” is also obfuscated by the same energetic electrons the closer one measures near the axis. This is another reason the emissive probe array (Sec. II B 2) is preferred. Nevertheless, Eq. (3) does provide useful plasma information in regions where energetic electrons do not have a major impact.

Langmuir probe measurements of ion density, plasma potential, and electron temperature as a function of radial distance from the chamber axis ($r = 0$) are shown in Fig. 6. For reference, the radius of the chamber wall is $r = 61$ cm. Error bars reflect uncertainties from the SMU source voltage, current collection, probe/sheath size, and the discretized nature of the measurements.³⁶ As the magnetic field and/or the emission current is increased, the ion density and plasma potential also increased inside the VCL volume ($r < 7.5$ cm) with on-axis densities and potentials up to 10^{13} cm⁻³ and -92 V, respectively. The on-axis plasma potential is positively correlated with the cathode emission voltage recorded from the power supply and can be used to estimate the VCL plasma potential if the keeper voltage is also known. Inside the rotational volume ($r > 7.5$ cm), the increase in the magnetic field and/or the emission current is associated with an increase in ion density. The plasma potential is indistinguishable except for the $B_{max} = 189$ G and $I_c = 3$ A case.

Electron heating is associated with an increase in emission current. Predictably, the electron temperature was measured to be several eV hotter in the VCL region than in the rotational volume. The mean temperatures in the VCL ($r \leq 7.5$ cm) are $T_e = 12.4$ eV for $I_c = 12$ A and $T_e = 8.3$ eV for $I_c = 3$ A. Inside the plasma rotation volume (7.5 cm $< r$), the mean temperatures are $T_e = 8.6$ eV for $I_c = 12$ A and $T_e = 7.0$ eV for $I_c = 3$ A. No data could be collected on-axis for the $B_{max} = 189$ G, $I_c = 12$ A case due to I-V curve truncation.

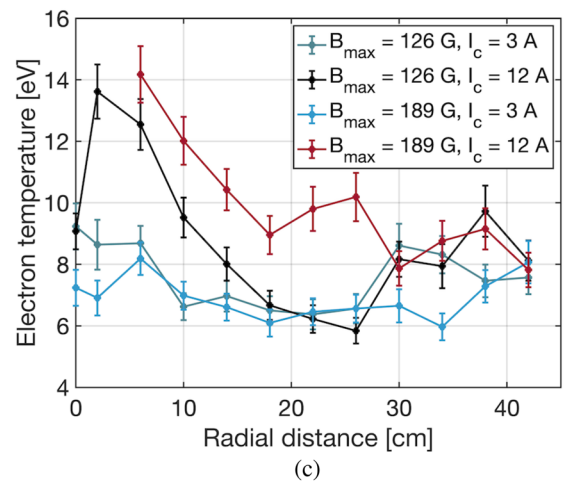
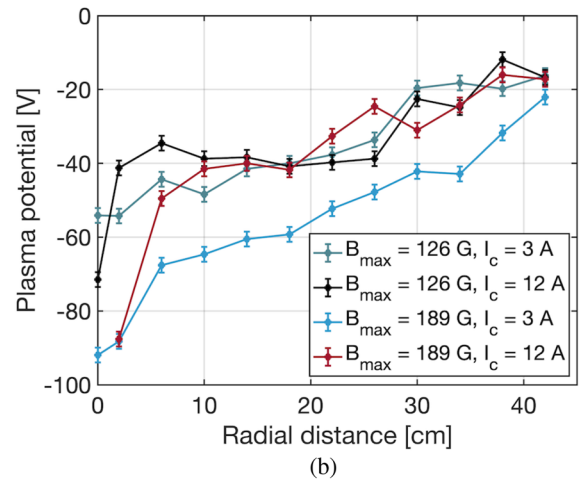
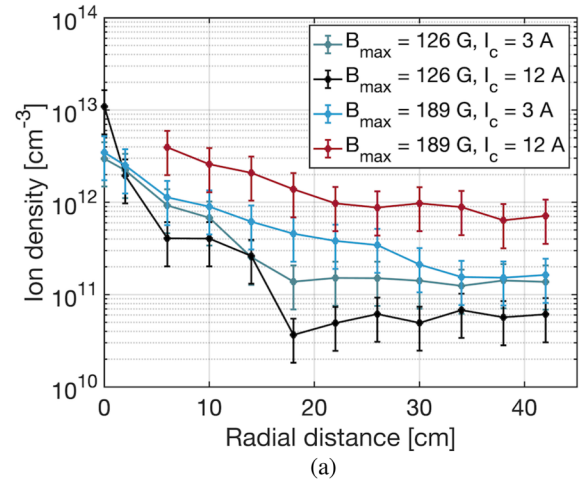


FIG. 6. (a) Langmuir probe ion density, (b) plasma potential, and (c) electron temperature profiles as a function of radial distance from the axis through the chamber midplane for two magnetic field strengths and HC emission currents. The chamber wall is at $r_w = 61$ cm.

2. Emissive probe array

There are several emissive probe analytical techniques available to implement³⁷ on EMPD, but the floating-point method is utilized because of its simplicity and short measurement time. In this technique, each probe is heated gradually by applying a current until its floating voltage approaches the lower edge of saturation. At sufficiently high heating currents (i.e., high electron emission currents), the sheath potential will saturate at $\approx -T_e/e$ relative to the plasma potential. This method is recommended for use in plasma environments with electron densities in the range³⁸ of 10^5 – 10^{12} cm^{-3} . At higher densities, the filament temperature required to achieve thermionic emission for the floating-point method will destroy the filament. At lower electron densities, space charge effects become non-negligible. When used in the proper regime, the precision with this method can be as low as $T_w/e = 0.2$ V,³⁷ where T_w is the temperature of the filament. Furthermore, the emissive probe measurements are expected to be sufficiently close to the plasma potential without excessive filament heating by orienting the probes perpendicular to the magnetic field. This has been experimentally supported up to at least 1010 G³⁹ and was found to hold in the presence of energetic electrons. Unlike the non-emissive probe that must charge (negative) to the energies of the electron energy distribution function (EEDF) tail to maintain current balance, the emissive probe charges more negative with regard to EEDF of the hot filament, which is generally higher in density and lower in temperature; thus only probe voltages on the order of a volt are required to balance 10s or 100s of eV tail electrons in the plasma, as long as validity conditions are met, such as the temperature being “small and constant.”^{37,40}

This durability across several orders of magnitude in density and magnetic field strength highlights why emissive probe plasma potential measurements are generally preferred over Langmuir probe measurements. Therefore, the emissive probe potential data are presented in Table I as the measurement of the plasma potential (V_p).

An emissive probe array comprising three probes was fabricated using two-bore alumina shafts between 0.3 and 0.6 m in length. Each emissive probe uses 0.10 mm OD thoriated tungsten filaments formed into a hairpin loop. Transformer wires were fed up the alumina shafts and used to form a compression fit with each filament. Approximately 2 cm of insulation at the ends of the transformer wire was removed to ensure electrical contact with the filaments. The probes are rotated on the midplane transverse to the magnetic field using a stepper motor and mounted on a translation stage aligned parallel to the chamber central axis. The probe array rotational axis has a 0.27 m horizontal and 0.32 m vertical displacement from the chamber axis. The probes intentionally avoid traversing through the VCL axis to reduce damage to the tungsten filaments.

The probe array is calibrated before each set of measurements by varying the floating potential as a function of heating current for each probe. The floating voltages typically saturate between 1.8 and 2.2 A of applied heating current and change over the lifetime of the filaments.

Four plasma potential cross sections were taken on the midplane perpendicular to the chamber axis for the same conditions as the Langmuir probe data. Two of the cases are shown in Fig. 7, each with a cathode current $I_c = 12$ A and an argon gas flow rate of

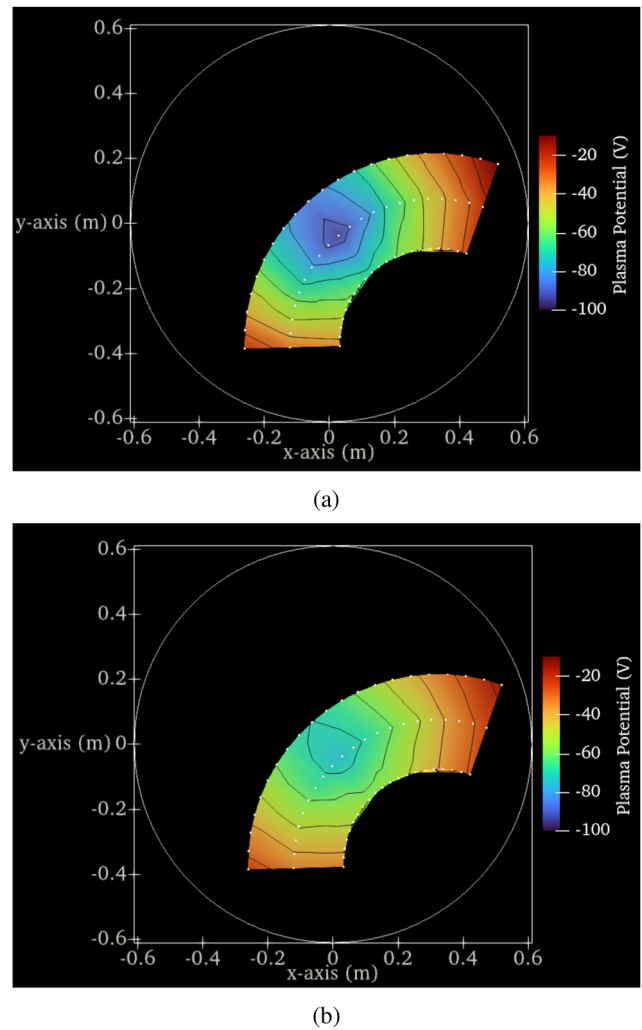


FIG. 7. Emissive probe plasma potential midplane cross sections for (a) $B_{\text{max}} = 189$ G and (b) $B_{\text{max}} = 126$ G. The HC emission current is $I_c = 12$ A for both cases. Surfaces are comprised of 63 measurements, and each contour represents a 10 V difference in potential.

20 SCCM. The surface plots are comprised of 63 emissive probe measurements fit to a 2D Delaunay mesh. A radial slice from the origin of the surface plot, for all four plasma environments, gives the plasma potential as a function of radial distance, as shown in Fig. 8. The plasma potential increases with magnetic field and emission current in the region $r < 30$ cm. For $r > 30$ cm, the plasma potentials all fall within 10 V of one another. The potential gradient is approximately constant in the rotation volume after about $r = 10$ cm, indicating a constant radial electric field. There is also a small potential gradient near the axis as one would expect if the VCL is behaving as an equipotential.

3. Retarding Mach probe

A twin retarding potential analyzer (RPA) has been designed to measure the azimuthal ion rotational velocity (v_ϕ) by collecting

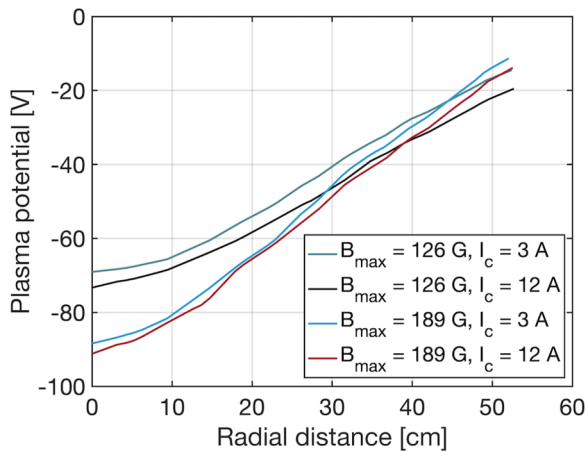


FIG. 8. Emissive probe plasma potential profiles as a function of radial distance from the axis. Curves are interpolated from the origin of the midplane surface mesh for varying magnetic fields and HC emission currents. The radius of the wall is $r_w = 61 \text{ cm}$.

the ion saturation current from opposite directions, thus simultaneously acting as a Mach probe. The diagnostic is referred to as the “retarding Mach probe” (RMP) shown in Fig. 9 and consists of four electroformed nickel mesh grids, a collecting disk, and a protective Macor plug. The housing material is made of Teflon, which is 13 mm wide/tall and 38 mm long. From the plasma facing grid

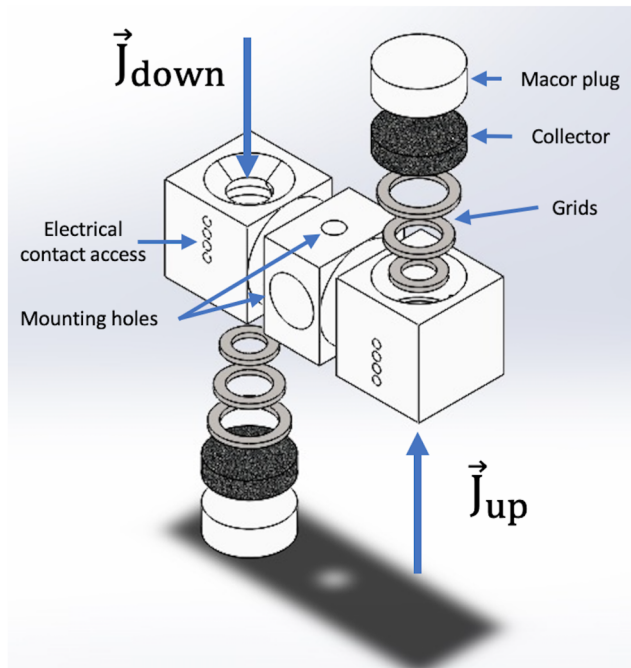


FIG. 9. Exploded view of the RMP showing the various grids, collector and protective Macor plug. The direction of collected ion current densities are indicated by $\vec{J}_{up}$ and $\vec{J}_{down}$.

working inward, there is a guard grid, electron repulsion grid, ion discriminating grid, and a secondary repulsion grid. All grids have a separation distance of 1.57 mm. The collecting disk sits between the secondary repulsion grid and a Macor plug. Four small electrical access holes are on the side of the Teflon housing, which also has a larger central mounting hole used to secure it to the radial Langmuir probe shaft. The plasma entrance holes are countersunk and have a bore radius of $r_p = 0.25 \text{ cm}$, much smaller than the ion Larmor radius ($r_p/r_{Li} \ll 1$), such that the ions are considered to be unmagnetized.

The collector is biased relative to chamber ground, while the grids are biased with respect to the collector. The discriminator is swept through a voltage range spanning -80 – 20 V , and the resulting ion current is a function of particle energy. The electron repulsion grids are typically set between -60 and -100 V .

There are two methods to determine the azimuthal ion flow velocity with the RMP and both assume a flowing. The first method is by taking the ratio of the upstream ion current density J_{up} and the downstream ion current density J_{down} in the ion saturation regions to determine the Mach number $M = v_\phi/c_s$. The current ratio for a 1D unmagnetized drifting Maxwell-Boltzmann distribution with the ion flux flowing parallel to a planar surface normal vector may be expressed as⁴¹

$$\frac{J_{up}}{J_{down}} = \exp(KM), \quad (4)$$

where the constant of proportionality $K = 1.34$ was chosen²⁶ for the case of unmagnetized ions and $T_i \leq 3T_e$. Then, the azimuthal ion flow velocity is given by

$$v_\phi = \frac{c_s}{K} \ln\left(\frac{J_{up}}{J_{down}}\right), \quad (5)$$

where the ion sound speed is $c_s = \sqrt{k_B(T_e + 3T_i)/m_i}$.

The second method to determine the ion flow velocity is from the ion velocity distribution function (IVDF) in each direction. The IVDF is again assumed to be described by a 1D drifting Maxwell-Boltzmann distribution with the most probable flow velocity at the maximum of the distribution and a width that is related to the ion temperature T_i . The IDVF can be shown⁴² to follow the following equation:

$$\frac{dI_{col}}{dV_d} \propto \exp\left(-\frac{m_i(v - v_\phi)^2}{2k_B T_i}\right), \quad (6)$$

where dI_{col}/dV_d is the derivative of the collector current with respect to the discriminator voltage.

A least squares fit to the IVDF given by Eq. (6) yields the ion temperature for either the upstream or downstream directions. The midpoint between maximum values of the IVDF fits corresponds to the discriminator voltage for no net flow. The voltage difference between this midpoint and either of the IVDF maximums is then taken to be the mean flow velocity (converted from energy in eV).

Currently, only the first method for determining the ion flow velocity [Eq. (5)] is employed because the IVDF fit associated with the downstream current cannot consistently be resolved for all plasma environments.

The RMP is mounted on the radial Langmuir probe alumina shaft 44.5 cm from the axis near the edge of the outer rotational

TABLE II. Several RMP quantities at radial location $r = 44.5$ cm for two cases of different magnetic field strengths and $I_c = 12$ A for both cases.

Parameters	$B_{\max} = 126$ G	$B_{\max} = 189$ G
$J_{\text{up}}/J_{\text{down}}$	3.98 ± 0.47	3.11 ± 0.24
M	1.03 ± 0.09	0.85 ± 0.06
T_e (eV)	8.12 ± 0.40	7.82 ± 0.26
T_i (eV)	1.20 ± 0.24	1.55 ± 0.29
c_s (km/s)	5.32 ± 0.19	5.49 ± 0.20
V_p (V)	-30.1 ± 7.0	-22.4 ± 7.5
v_ϕ (km/s)	5.27 ± 0.66	4.65 ± 0.48
E_r (V/m)	-117	-163
B_z (G)	71.9	107.9
E_r/B_z (km/s)	16.3	15.1

volume for two of the four previous cases, $B_{\max} = 126$ G and $B_{\max} = 189$ G, where $I_c = 12$ A for both. The plasma parameters at this location are presented in Table II. Here, T_e is determined by using the radial Langmuir probe. Both T_i and V_p are found from a least squares fit to Eq. (6) on the upstream IVDF. The ion temperature is between 1 and 2 eV in this region, and the plasma potential is -30 and -22 V for the $B = 126$ G case and $B = 189$ G case, respectively. These potentials are in agreement with the emissive probe measurements to within a few volts. The radial electric field E_r is given by the negative gradient of the linear region of the plasma potential measured by the emissive probe array shown in Fig. 8. The axial magnetic field B_z is estimated by a FEM calculation at the RMP location. The $E \times B$ drift velocity is given by the ratio E_r/B_z (see discussion in Sec. III). RMP data and the calculated quantities for the two cases are presented in Table II. The largest rotational velocity was found to be 5.27 km/s, and increasing the magnetic field yields a value of 4.65 km/s.

The proportionality constant K in Eq. (5) is a function of the probe geometry and should be determined experimentally against other diagnostics or determined from a particle-in-cell (PIC) model containing the probe geometry. The RMP is designed to experimentally measure K . This is done by using method two and requires that the downstream IVDF is resolved. If both IVDFs are available, then the rotational velocity can be independently determined by finding the midpoint between the upstream and downstream Gaussian centroids. One can then calculate the proportionality constant by using Eq. (4), assuming the same current ratio from method one. It should be noted that K is usually also a function of the plasma parameters. Even so, this experimentally determined value can be assumed to be constant in similar plasma environments when the IVDF(s) cannot be resolved.

The importance of K is highlighted by using method two. The only partially resolved downstream IVDF in the dataset was for the $I_c = 12$ A and $B = 189$ G case. Using the midpoint between the mean values of the two distributions yields a mean ion drift energy of 9.1 eV. This gives a mean (95% confidence) ion flow velocity of $v_\phi = 6.6 \pm 0.85$ km/s and a value of $K = 0.94$ by Eq. (5). In contrast, if the method to estimate K used by Hudis and Lidsky⁴³ is implemented, $K = 1.5$ given the values presented in Table II.

III. RESULTS

The plasma environment for the low-field low-current settings is over-dense ($f_{pe}/f_{ce} > 1$) with ion densities and electron temperatures throughout the rotational volume between 10^{11} – 10^{12} cm⁻³ and 6–9 eV, respectively. Ion temperatures were between 1 and 2 eV on the outer edge with rotational velocities between 4.6 and 5.3 km/s. The plasma is moderately collisional as the ion–neutral and ion–ion collision frequencies can be significant.

There are two means to increase the ion density. The first is to increase the voltage to the cathode. The emission current quickly increases and is associated with hotter electrons, a slight rise to the plasma potential, and larger ion density. Since the plasma potential is relatively unchanged, the enhanced density in this case is attributed to an increase in the rate of electron impact ionization.

The second method to increase the ion density is to apply a stronger magnetic field. This causes the electric field to increase due to the reduced electron mobility transverse to the magnetic field. The stronger electric field and a reduced Larmor radius increase the ion confinement time, thereby increasing the ion density while keeping the mirror ratio constant. The applied magnetic field has also shown to provide the most direct influence over the plasma potential.

The emissive probe array indicated a linear potential in the midplane rotational volume, a near equipotential in the outer VCL region, and a small increase in plasma potential with an increase in the cathode emission current. At times, the Langmuir probes gave potential measurements that were up to 33% less than the emissive probes. The longest and middle length emissive probes of the array split the VCL axis to reduce damage to the probe filaments and ceramic bonding agent. This has the benefit of increasing the lifetime of the probe and reducing error. However, if the VCL was not an equipotential, this may have gone undetected in the cross sections shown in Fig. 8. Lifetime-improving modifications to the probes are needed to determine the potential in the VCL region. As mentioned previously, the emissive probe analysis using the floating point method may produce measurements as much as T_e/e below the plasma potential. Electron heating was shown to occur near the chamber axis when the cathode current is increased compared to the outer region near the wall of a few to several eV. This may imply a slightly reduced radial electric field for the $I_c = 12$ A cases.

Currently, the RMP only measures ion flow velocity in the outer region of the plasma rotational volume where the Debye length is on the order of the internal RMP grid spacing. In the denser regions, the RMP is susceptible to space-charge buildup between the grids, influencing the measured saturation current. The preliminary rotational velocity measurements do not show a statistically significant correlation to the velocity E_r/B_z . The measured rotational velocities are also lower than both the argon CIV limit (8.7 km/s) and the E_r/B_z velocities. This is most likely due to large Larmor effects (LLEs) present in the unmagnetized ion population. In a spatially nonuniform electric field, the $E \times B$ drift velocity has additional higher terms. The second-order term scales linearly with the expansion parameter⁴⁴ $\varepsilon = r_{Li}/L_\perp$, where $L_\perp$ is the spatial gradient length scale of the transverse electric field. Evaluation at the radial location of the RMP yields $L_\perp \sim |\nabla E_r/E_r| \sim 1$ (i.e., about the same length scale of the chamber radius) for all four electric fields shown in Fig. 8. The ion Larmor radius is $r_{Li} = 30$ cm and $r_{Li} = 55$ cm given T_i and B_z in Table II. Therefore, the ion Larmor radius is comparable with the field

gradient length scale ($\epsilon \sim 1$), and the second-order drift term and LLE are not negligible for both RMP measurements. This is important as LLE are known to cause particle trajectories to exhibit stochastic behavior across magnetic field lines and may significantly influence the $E \times B$ drift velocity throughout the entire plasma length scale. The second-order term to the drift velocity $\sim r_{Li}^2 \nabla^2 E_r / B_z$ was not included in the analysis in Sec. II B 3 as the spatial resolution of the plasma potential was too low. The dependence of the drift velocity on the ion Larmor radius should be carefully accounted for in any future studies at these field strengths.

The results presented here have not taken into account secondary electron emission (SEE) or sputtered copper buildup on the borosilicate glass walls, which periodically require cleaning. These effects are expected to be minimal but influence the electric field boundary conditions if left to accumulate. The C-based hollow cathode sputtering causes a thin layer of carbon deposition on the inside of the entire chamber. The periodic cleaning that this requires also removes any sputtered copper. Since these measurements, the Cu plates have been graphite-coated.

IV. CONCLUSIONS

A quasi-stationary rotating plasma device in a magnetic mirror configuration has been developed at AFRL, Kirtland Air Force Base, to study dynamic plasma coupling in near-Earth space plasma. The $E \times B$ magnetized plasma device uses a steady state axial hollow cathode source that produces a plasma discharge in electrical contact with a floating conductor. The discharge forms an axisymmetric equipotential surface called the Virtual Cathode Lightsaber, which sustains a rotating plasma using the crossed field technique.

The measured plasma parameters were reported along with several electrostatic diagnostics. The analytical methods used to perform a basic characterization of a flowing magnetized plasma were described. The measurements and utilized methods are benchmarks for the low magnetic field and low cathode current regime at the chamber midplane in EMPD. The ability to control the plasma density and, to an extent, the plasma potential by the applied magnetic field and cathode voltage was demonstrated along with the presence of plasma rotation in the outer rotational volume.

One significant structural change to the vacuum chamber and the poppet valve leading to the cryopump will be replaced by a flush mounted gate valve that is expected to halve the chamber operating pressure.

Two additional diagnostics are under construction. The first is the introduction of a plasma impedance (resonance) probe to provide electron density measurements by a direct measurement of the plasma frequency. This would negate the requirement to assume the form of the electron distribution function inherent in Langmuir probe analysis. In addition, laser-induced fluorescence (LIF) measurements using a dye laser will provide Doppler shift information by targeting a two-photon transition of argon. The anticipated LIF signal intensity is sufficient to resolve rotational velocity, ion temperatures, and diffusion coefficients.⁴⁵

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Charles T. Hooper: Conceptualization (equal); Data curation (supporting); Formal analysis (supporting); Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jenny R. Smith:** Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Trenton R. Brewer:** Data curation (supporting); Investigation (supporting); Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Jonathon R. Heinrich:** Conceptualization (supporting); Investigation (supporting); Methodology (supporting). **Remington Reid:** Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Visualization (supporting); Writing – review & editing (supporting). **David L. Cooke:** Conceptualization (lead); Data curation (supporting); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (supporting); Project administration (lead); Writing – original draft (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author, J. R. Smith, upon reasonable request.

REFERENCES

- ¹C. K. Goertz and R. W. Boswell, “Magnetosphere-ionosphere coupling,” *J. Geophys. Res.: Space Phys.* **84**, 7239–7246, <https://doi.org/10.1029/ja084ia12p07239> (1979).
- ²H. Alfvén, “Collision between a nonionized gas and a magnetized plasma,” *Rev. Mod. Phys.* **32**, 710–713 (1960).
- ³K. Liou, R. B. Torbert, and G. Haerendel, “Momentum coupling in the ‘CRIT II’ critical ionization velocity experiment,” *J. Geophys. Res.: Space Phys.* **101**, 19649–19657, <https://doi.org/10.1029/96JA00393> (1996).
- ⁴D. M. Ober, T. S. Crawford, J. V. Eccles, and J. M. Holmes, “3D multi-fluid MHD simulation of the early time behavior of an artificial plasma cloud in the bottom side ionosphere,” *J. Geophys. Res.: Space Phys.* **126**, e2020JA029036, <https://doi.org/10.1029/2020ja029036> (2021).
- ⁵R. F. Ellis, A. B. Hassam, S. Messer, and B. R. Osborn, “An experiment to test centrifugal confinement for fusion,” *Phys. Plasmas* **8**, 2057–2065 (2001).

- ⁶D. L. Cooke and I. Katz, "TSS-1R electron currents: Magnetic limited collection from a heated presheath," *Geophys. Res. Lett.* **25**, 753–756, <https://doi.org/10.1029/97gl03102> (1998).
- ⁷R. G. D'Arcy and S. A. Colgate, "Measurements at the southern magnetic conjugate region of the fission debris from the starfish nuclear detonation," *J. Geophys. Res.* **70**, 3147–3159, <https://doi.org/10.1029/JZ070i013p03147> (1965).
- ⁸P. Delamere, H. Stenbaek-Nielsen, D. Hampton, and E. Wescott, "Optical observations of the early ($t < 5$ s) ion dynamics of the CRRES G1, G9, and G11A releases," *J. Geophys. Res.: Space Phys.* **101**, 17243–17257, <https://doi.org/10.1029/96JA00862> (1996).
- ⁹B. Lehnert, "Rotating plasmas," *Nucl. Fusion* **11**, 485 (1971).
- ¹⁰D. Baker, J. Hammel, and F. Ribe, "Rotating plasma experiments. I. Hydromagnetic properties," *Phys. Fluids* **4**, 1534–1548 (1961).
- ¹¹D. Baker and J. Hammel, "Rotating plasma experiments. II. Energy measurements and the velocity limiting effect," *Phys. Fluids* **4**, 1549–1558 (1961).
- ¹²M. Bennett, S. Lebedev, G. Hall, L. Suttle, G. Burdiak, F. Suzuki-Vidal, J. Hare, G. Swadling, S. Patankar, M. Bocchi, J. Chittenden, R. Smith, A. Frank, E. Blackman, R. Drake, and A. Ciardi, "Formation of radiatively cooled, supersonically rotating, plasma flows in Z-pinch experiments: Towards the development of an experimental platform to study accretion disk physics in the laboratory," *High Energy Density Phys.* **17**, 63–67 (2015), part of Special Issue: 10th International Conference on High Energy Density Laboratory Astrophysics.
- ¹³A. Ando, H. Tobari, M. Shibata, H. Isobe, K. Hattori, M. Inutake, K. Nemoto, H. Higaki, and M. Ichimura, "Plasma flow measurement by Mach probes in GAMMA 10," *Fusion Sci. Technol.* **51**, 217–219 (2007).
- ¹⁴R. Reid, C. Romero-Talamás, W. Young, R. Ellis, and A. Hassam, "100 eV electron temperatures in the Maryland centrifugal experiment observed using electron Bernstein emission," *Phys. Plasmas* **21**, 063305 (2014).
- ¹⁵R. Ellis, A. Case, R. Elton, J. Ghosh, H. Griem, A. Hassam, R. Lunsford, S. Messer, and C. Teodorescu, "Steady supersonically rotating plasmas in the Maryland centrifugal experiment," *Phys. Plasmas* **12**, 055704 (2005).
- ¹⁶R. Reid, Personal correspondence, 2023.
- ¹⁷J. Y. Kim, J. Y. Jang, J. Choi, J. i. Wang, W. I. Jeong, M. A. I. Elgarhy, G. Go, K.-J. Chung, and Y. S. Hwang, "Magnetic confinement and instability in partially magnetized plasma," *Plasma Sources Sci. Technol.* **30**, 025011 (2021).
- ¹⁸C. A. Romero-Talamás, R. C. Elton, W. C. Young, R. Reid, and R. F. Ellis, "Isorotation and differential rotation in a magnetic mirror with imposed $E \times B$ rotation," *Phys. Plasmas* **19**, 072501 (2012).
- ¹⁹K. Xie, R. A. Martinez, and J. D. Williams, "Current-voltage characteristics of a cathodic plasma contactor with discharge chamber for application in electrodynamic tether propulsion," *J. Phys. D: Appl. Phys.* **47**, 155501 (2014).
- ²⁰K. Xie, Q. Xia, J. D. Williams, R. A. Martinez, and C. C. Farnell, "Extracted current, bias voltage, and ion production of cathodic hollow-cathode-driven plasma contactors," *J. Spacecr. Rockets* **52**, 1181–1192 (2015).
- ²¹J. D. Williams and P. J. Wilbur, "Electron emission from a hollow cathode-based plasma contactor," *J. Spacecr. Rockets* **29**, 820–829 (1992).
- ²²D. L. Morse, "Plasma rotation in a hollow-cathode discharge," *Phys. Fluids* **8**, 516–521 (1965).
- ²³S. T. Lai, "A review of critical ionization velocity," *Rev. Geophys.* **39**, 471–506, <https://doi.org/10.1029/2000rg000087> (2001).
- ²⁴C. K. Goertz, S. Machida, and R. A. Smith, "An asymptotic state of the critical ionization velocity phenomenon," *J. Geophys. Res.: Space Phys.* **90**, 12230–12234, <https://doi.org/10.1029/JA090iA12p12230> (1985).
- ²⁵F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion* (Springer International Publishing, 2016).
- ²⁶I. H. Hutchinson, *Principles of Plasma Diagnostics*, 2nd ed. (Cambridge University Press, 2002).
- ²⁷H. Gunell, M. Larsson, and N. Brenning, "Experiments on anomalous electron currents to a positive probe in a magnetized plasma stream," *Geophys. Res. Lett.* **27**, 161–164, <https://doi.org/10.1029/1999gl010694> (2000).
- ²⁸N. H. Stone and C. Bonifazi, "The TSS-1R mission: Overview and scientific context," *Geophys. Res. Lett.* **25**, 409–412, <https://doi.org/10.1029/97gl02980> (1998).
- ²⁹J. R. Heinrich and D. L. Cooke, "Dynamics of positive probes in underdense, strongly magnetized, $E \times B$ drifting plasma: Particle-in-cell simulations," *Phys. Plasmas* **20**, 093503 (2013).
- ³⁰D. C. Meeker, *Finite element method magnetics, version 4.2 (20 April 2019 build)*, <https://www.femm.info>.
- ³¹D. M. Goebel, G. Becatti, I. G. Mikellides, and A. Lopez Ortega, "Plasma hollow cathodes," *J. Appl. Phys.* **130**, 050902 (2021).
- ³²V. Vekselman, Y. E. Krasik, S. Gleizer, V. T. Gurovich, A. Warshavsky, and L. Rabinovich, "Characterization of a heaterless hollow cathode," *J. Propul. Power* **29**, 475–486 (2013).
- ³³D. Lev, G. Alon, and L. Appel, "Low current heaterless hollow cathode neutralizer for plasma propulsion—Development overview," *Rev. Sci. Instrum.* **90**, 113303 (2019).
- ³⁴K. i. Maeda, "Conductivity and drifts in the ionosphere," *J. Atmos. Terr. Phys.* **39**, 1041–1053 (1977).
- ³⁵Impedans, Langmuir Probe Installation and User Guide, Impedans Ltd..
- ³⁶R. B. Lobbia and B. E. Beal, "Recommended practice for use of Langmuir probes in electric propulsion testing," *J. Propul. Power* **33**, 566 (2017).
- ³⁷J. P. Sheehan and N. Hershkowitz, "Emissive probes," *Plasma Sources Sci. Technol.* **20**, 063001 (2011).
- ³⁸R. F. Kemp and J. M. Sellen, "Plasma potential measurements by electron emissive probes," *Rev. Sci. Instrum.* **37**, 455–461 (1966).
- ³⁹H. Fujita, S. Nowak, B. Hoegger, and H. Schneider, "Potential measurements by an emissive probe in a magnetized plasma," *Phys. Lett. A* **78**, 263–265 (1980).
- ⁴⁰N. Hershkowitz and M. Cho, "Measurement of plasma potential using collecting and emitting probes," *J. Vac. Sci. Technol. A* **6**, 2054–2059 (1988).
- ⁴¹K.-S. Chung, "Why is the Mach probe formula expressed as $R = J_{up}/J_{dn} = \exp[KM_{\infty}]?$," *Jpn. J. Appl. Phys.* **45**, 7914 (2006).
- ⁴²C. Böhm and J. Perrin, "Retarding-field analyzer for measurements of ion energy distributions and secondary electron emission coefficients in low-pressure radio frequency discharges," *Rev. Sci. Instrum.* **64**, 31–44 (1993); https://pubs.aip.org/aip/rsi/article-pdf/64/1/31/19184690/31_1_online.pdf.
- ⁴³M. Hudis and L. M. Lidsky, "Directional Langmuir probe," *J. Appl. Phys.* **41**, 5011–5017 (1970).
- ⁴⁴K. Stasiewicz, "Finite Larmor radius effects in the magnetosphere," *Space Sci. Rev.* **65**, 221–252 (1993).
- ⁴⁵H. Matsuta and K. Kitagawa, "Two-photon laser-induced fluorescence spectra of argon in a Grimm-style glow discharge tube," *Spectrosc. Lett.* **45**, 13–16 (2012).